COMBINED SCIENCE

FORM 1-2

PHYSICS NOTES

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TOPICS

- ❖ **DATA PRESENTATION**
- ❖ **MEASUREMENTS**
- ❖ **FORCE**
- ❖ **MECHANICAL SYSTEMS**
- ❖ **ENERGY**
- ❖ **MAGNETISM**
- ❖ **ELECTRICITY**

DATA PRESENTATION

WAYS OF PRESENTING DATA

- Data is any information such as facts and statistics gathered by scientist during experiment or during a research.
- Scientists interpret data (i.e. understand and explain the meaning of) and it to make conclusions about their experiment.
- Data gathered need to be presented in a certain way which is not time consuming e.g. visual or graphic.
- Data can be qualitative data (observations) or qualitative data (statistical data).

Tallies

- Tally marks are a unary numeral system. They are a form of numeral used for counting. They are most useful in counting or tallying ongoing results, such as the score in a game or sport, as no intermediate results need to be erased or discarded.
- However, because of the length of large numbers, tallies are not commonly used for static text.
- Tally marks are typically clustered in groups of five for legibility. The cluster size 5 has the advantages of easy conversion into decimal for higher arithmetic operations and avoiding error, as humans can far more easily correctly identify a cluster of 5 than one of 10.
- Tallies are done as follows I II III IIII ~~IIII~~

A tally of mass of form one learners

Mass (kg)	Number of learners	tally
31 – 35	1	I
36 – 40	6	~~IIII~~ I
41 – 45	11	~~IIII~~ ~~IIII~~ I
46 – 50	8	~~IIII~~ III
51 - 55	4	1111

Tables

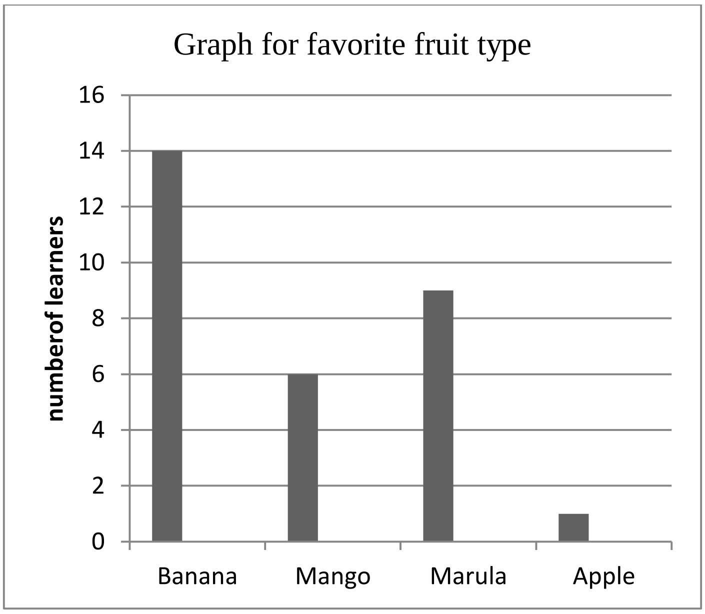
- Presenting data in a table helps to make it clear and easy to read and understand.
- Tables should always have a descriptive heading
- Each column and row should be labelled and can have a unit of measurement representing all data on the column if applicable. For example;

Favourite fruit types for form one learners

Types of fruit	Number of learners
Banana	14
Mango	6
Marula	9
Apple	1

Bar graphs

- A bar graph is a visual display of data on a graph. Bar graphs have vertical bars of different heights on a pair of axes.
- Bar graphs are used when the data is in groups or categories e.g. days of the week, types of transport, type of fruits e.t.c.
- We can use the information presented in the table above to draw a bar graph

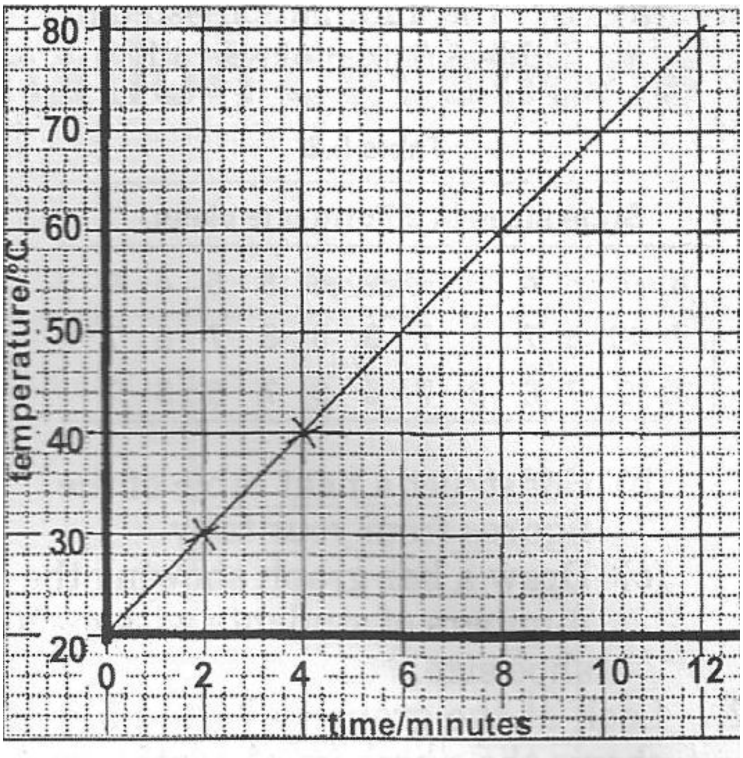


- The heights of the bars shows the values represented in the table
- From the graph we can see that bananas are the most popular fruit and apples are the least popular.

STRAIGHT LINE GRAPHS

- Shows a variable that change with time e.g. temperature change as water is heated,
- It has x-axis and y-axis
- Points to note when drawing a line graph include;
    - Draw the lines to show the x-axis and y-axis
    - Label the axis
    - Select suitable scale for both the x-axis and y-axis
    - Mark points accurately using the following mark ● or +
    - Draw a smooth line to join the points i.e. draw a line of best fit using a sharp pencil

A line graph showing temperature changes with time



- With such a straight line we can conclude that the temperature is directly proportional to the time. As time increases temperature also increase
- The graph can be used to find the temperature at 3 and 7 minutes

Examples 2 A student carried out an investigation the effect of height on pressure of a liquid. Table below shows results obtained from the experiment.

depth/cm	10	20	30	40	50
height/cm	0.5	1.0	1.5	1.8	2.5

a) Plot a graph of height against the depth

b) Find , using the graph, the height at depth of 24cm

c) Describe, from the graph, the relationship between height and depth

Example 3 An experiment was done to investigate the relationship between the distances travelled over the time taken for a car to move along a track. The following data was collected.

Time/s	0	2	4	6	8	10
Distance/m	0	5	10	15	20	25

(a) Plot a graph of distance travelled against time taken.

(b) What is the relationship between the variables

MEASUREMENT

PHYSICAL QUANTITIES

Estimating Quantities

- An estimate is a guess very close to actual based on knowledge or rough calculations and it can be done before actual measurement. Different people produce different estimates for a given quantity.
- Estimate the length of the following and record your estimations in metres
  (a) Width of the classroom door
  (b) The length of the classroom walls
  (c) Height of your desk
- Estimate and record estimates in kg / g of mass of the following
  (a) A science textbook
  (b) A pen
  (c) A beaker
- Estimate and record your estimations in ^{o}C of temperature of
  (a) Cold water
  (b) Warm water
  (c) Hot water
- Estimate and record your estimation in seconds or minutes of time it takes to
  (a) Walk the length of the class
  (b) Take your book out of your satchel and place it on your desk
  (c) To boil 100ml of water
- Record all observations in the table below

Item	Estimation	Actual Measurement	Accurate /not accurate

ERRORS IN MEASUREMENT

- Errors occur in all physical measurements and there are two common errors that could occur when taking measurements i.e.

Parallax error

- It is an error in reading an instrument due to the incorrect position of the eye.
- To avoid parallax error, the person taking measurement must make sure that their line of sight is directly in line with the instrument's pointer and scale.

Zero error

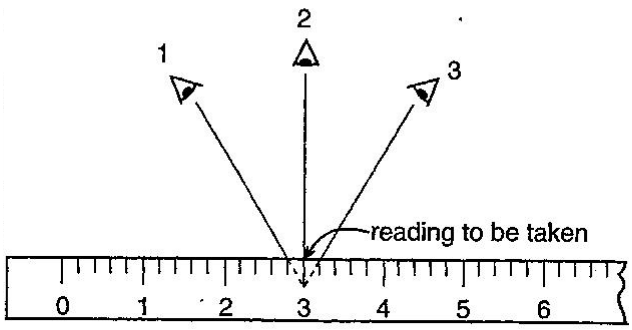
- Is caused by incorrect positioning of the zero point.
- The pointer on the instrument must be exactly positioned near to zero on the scale

PHYSICAL QUANTITIES AND SI UNITS

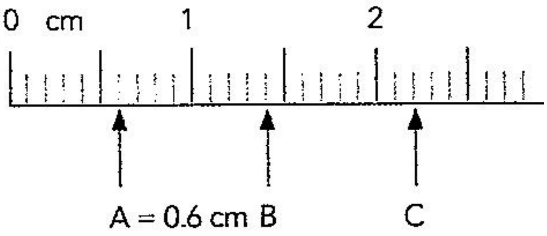
- Physical quantity is a property of an object or other substance that can be measured using an appropriate measuring instrument.
- Each type of measurement is done in a special unit, set as a standard to be used by scientists and other people so that they communicate effectively amongst themselves.
- SI units stand for **international system of units.**
- The International System (SI) units of measurements are used to measure physical quantities.

Reading an instrument scale

- It is important to be accurate when taking measurements.
- Errors in measurement can be minimised to ensure accurate measurement
- The eye must be correctly positioned in order to take correct readings
- In diagram below position 2 will give correct reading. Position 1 will give a reading that is less than the actual measurement and position 3 will give a reading that is greater than the actual measurement.



- To read the scale properly, count the number of divisions between zero and one. For example, in the scale below, there are ten divisions between zero and one. Divide 1 by the number of subdivisions to find out what each subdivision measures, in this case, each subdivision represents 0.1cm.



- Therefore reading at position A is 0.6cm; at position B is 1.4cm and position C is 2.2cm

Units including SI Units

Physical quantity	SI unit	Instruments
Length	Metre (m)	Metre rule, measuring tape
Mass	Kilogram (kg)	Balance
Time	Second (s)	Clock, stop watch
Temperature	Kelvin (K)	Thermometer

CONVERTING UNITS

Prefixes of SI units

- Kilo- (k) 1000
- Milli- (m) 0.001
- Centi- (c) 0.01

Length

- The SI unit is the metre and other units include centimetre (cm) and millimetres (mm).

  100cm = 1m

  10mm = 1cm
- When converting;

  (a) Metre to centimetre multiply metres by 100

(b) Centimetre to metres divide by 100

(c) Millimetre to centimetre divide by 10

(d) Centimetre to millimetre multiply by 10

Mass

- The SI unit is the kilogram and other units include the gram and the milligram

  1kg = 1000g

  1g = 0.001kg

- When converting;

  (a) Kilograms to grams multiply by 1000

  (b) Grams to kilograms divide by 1000

Time

- SI unit is the second. The other units include minutes and hours

  1h = 60mins

  1min = 60s

  1h = 360 s

- When converting;

  (a) Hour to minutes multiply by 60

  (b) Minutes to hours divide by 60

  (c) Seconds to minutes divide by 60

  (d) Minute to seconds multiply by 60

Temperature

- SI unit is the Kelvin. The other units include degrees Celsius (°C) and degrees Fareh (°F)

  0°C = 273K

- When converting;

  (a) Degrees Celsius to Kelvin add 273

  (b) Kelvin to degrees Celsius subtract 273

MEASURING PHYSICAL QUANTITIES

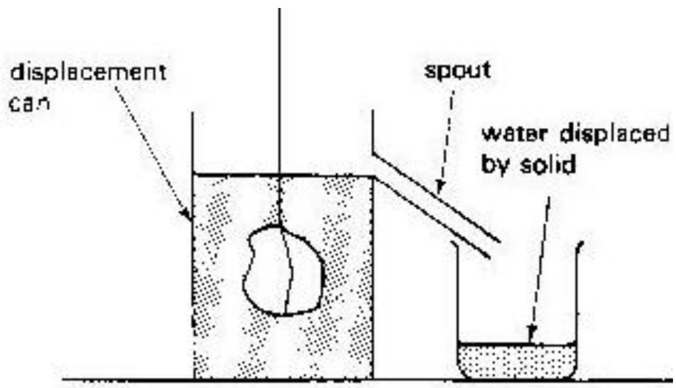
Measuring mass of liquid

- A liquid cannot be weighed directly without placing it in a container.
- The mass of an empty beaker is found on a balance.
- A known volume of the liquid is transferred from a measuring cylinder into the beaker. The mass of the beaker plus liquid is found.
- Therefore the mass of liquid is obtained by subtraction as follows.

$$\text{mass of water} = \text{mass of beaker and water} - \text{mass of empty container}$$

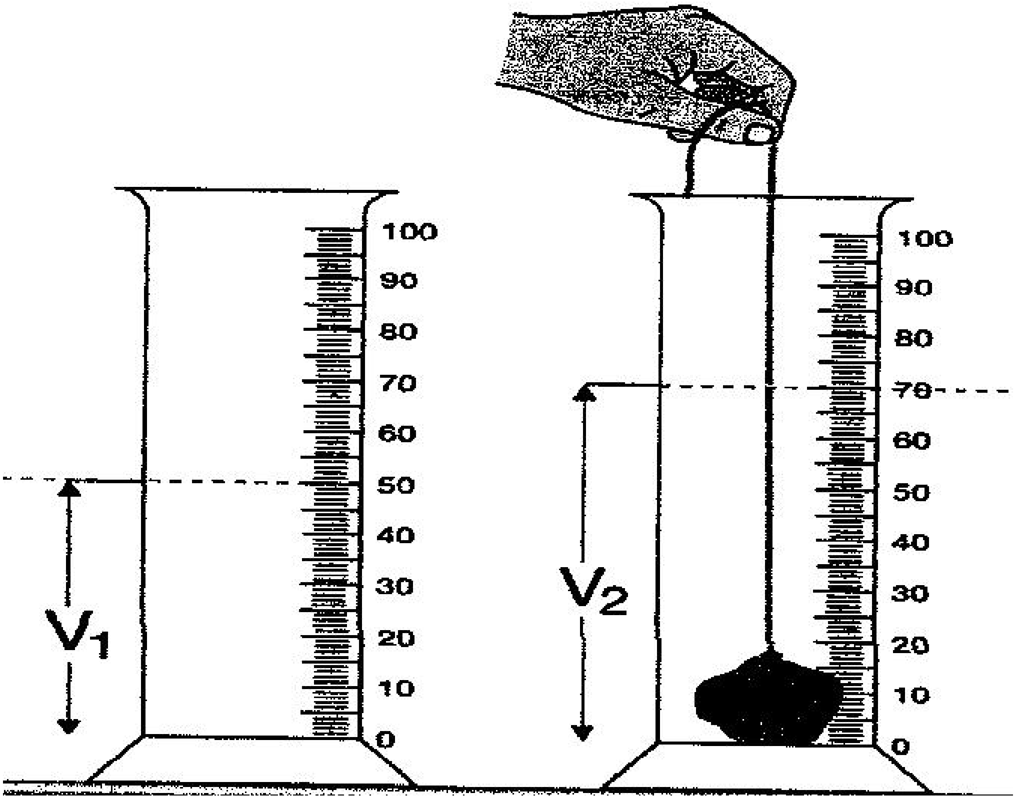
Volume of irregular objects

- Volume of an irregular object can be found by displacement method.
- When a solid is immersed in water, it displaces its own volume of liquid in which it is immersed.
- Fill the overflow can so that water is level with the bottom of the spout. Then place the object in the can, collecting the water which overflows.
- Measure its volume; this equals the volume of the object.



- If the displacement can is unavailable, use a measuring cylinder instead. Fill it about half-full with water; read and record the volume of water. (initial volume)
- Put the object in water so that it is completely covered; read and record the total volume of water and object (final volume)
- Calculate the volume of the object

$$volume\ of\ object = final\ volume - initial\ volume$$



Original volume of water = $50cm^3$

Final volume of water = $70cm^3$

Volume of stone $= 70cm^3 - 50cm^3 = 20cm^3$

Mass of small object

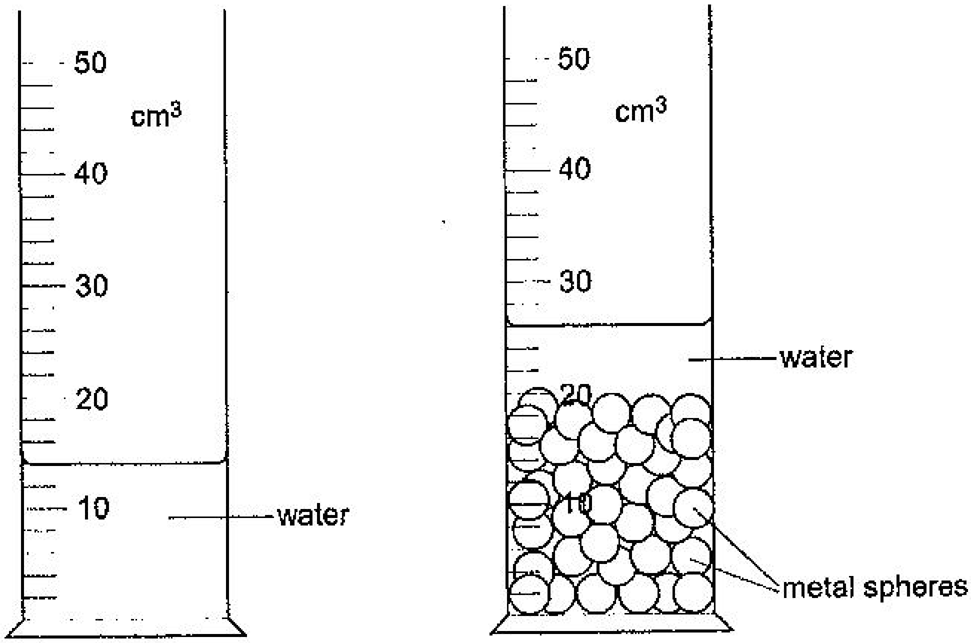
- If the object is very small, it is very difficult to measure its mass e.g. the mass of one bean seed or drawing pin.
- Measure the mass of a large number of the objects and then divide the mass by number of objects to get mass of one object

$$\text{mass of one seed} = \frac{\text{mass of seeds}}{\text{number of seeds}}$$

Volume of small objects

- Fill a measuring cylinder with water and record the volume.
- Place a large number of objects e.g. 50seeds into the water in the measuring cylinder. Record the new volume.

- Calculate the volume of the large number of the objects by subtracting the volume of water from that of water and seeds.
- Calculate the volume of one object by dividing the volume by number of objects



Volume of water = $14cm^3$

Volume of water + 50seeds = $26cm^3$

Volume of 50seeds = *26 – 14* = $12cm^3$

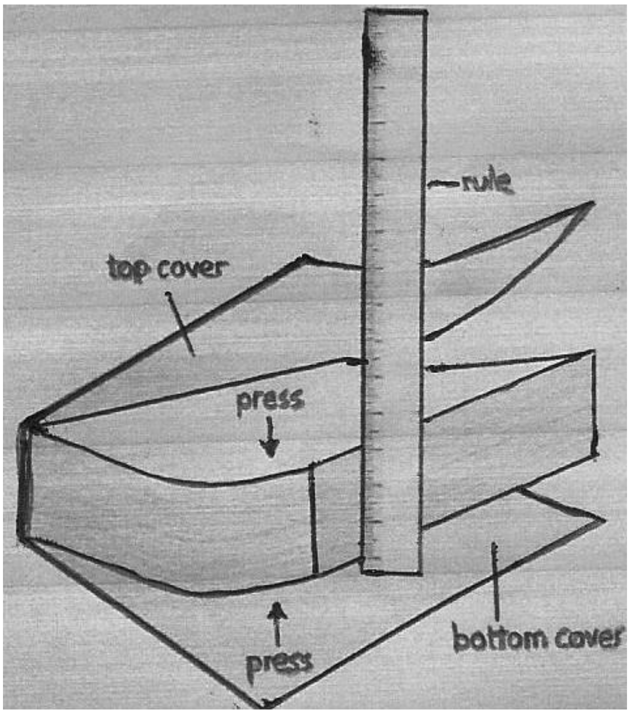
Volume of 1 seed = *12/50* = $2cm^3$

Thickness of small objects

- Separate the cover of the book from the rest of the pages
- Press together the pages and then count the number of sheets in your book.
- Measure the thickness of the sheets.
- Divide the thickness of the sheet by number of sheets to get thickness of one sheet.

$$thickness\ of\ one\ sheet = \frac{thickness\ of\ sheets}{number\ of\ sheets}$$

- Since the thickness of one sheet is so small you can convert cm to mm



Example 1 A pile of exercise books without covers is 20cm high. There are 30 exercise books each with 46 pages. What is the thickness of one sheet of paper?

Number of pages = 30 x 46 = 1380

Number of sheets = $\frac{1380}{2}$ = 690

Thickness of one sheet = $\frac{20cm\ x\ 10mm}{690}$ = $\frac{200}{690}$ =0,29mm

DENSITY

- Density is mass per unit volume of a substance

  Density (D) = $\frac{mass\ (m)}{volume\ (V)}$

- Units can be g/cm^3 or kg/m^3

Example 1 Calculate the density of glass if $120cm^3$ of glass has a mass of 300g

Example 2 A cylinder of aluminium has a radius of 7cm and a height of 20cm. The mass of the cylinder is 8.316kg. Calculate the density of aluminium

Example 3 A beaker has a mass of 48g. When $120cm^3$ of copper sulphate solution are poured into the beaker it is found to have a mass of 174g. Calculate the density of the copper sulphate

FORCE

EFFECTS OF FORCES

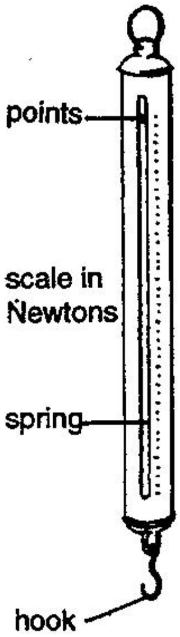
- A force is a push or a pull; a squeeze or a twist
- Effects of forces include;
    - Distortion or deformation (change in shape and size) – a force can change the shape of a solid object e.g. when you squeeze a cool drink can, the force exerted on the can by the hand causes the can to change shape.
    - Change in speed – a force can cause the speed of a moving object to change. It can cause the object to accelerate and move faster or decelerate and moves slower e.g. when riding a bicycle the harder and faster you pedal, the greater the force and the faster the bicycle goes.
    - Change direction – a force can cause a moving object to change direction e.g. a tennis ball approaching a tennis player, the player hits the ball and it changes direction because the racket exerts a force on the ball to change direction
    - Change in position – a force can cause a stationery object to move and change position e.g. if a soccer ball is kicked, the foot exert the force on the ball and the ball moves forward.

Types of force

- There are two main types of forces i.e. contact and non-contact forces
- Contact forces are forces that are in direct contact with each other e.g. pulling, pushing and twisting forces (mechanical forces) that are directly applied to an object.
    - Weight – is the force acting on mass due to gravity
    - Mechanical force – caused by the movement of objects e.g. falling water
    - Friction – force that opposes motion. It occurs between two objects that are moving against each other. It slows down a moving object or causes it to stop.
- Non contact forces that not in direct contact with each other. Effect of the force can be observed e.g.
    - Gravitational force – that pulls objects towards the centre of the earth.
    - Magnetic force – force exerted by magnets to attract or repel materials
    - Electrostatic force – produced by rubbing and can attract or repel objects

MEASURING FORCE

- Is measured in Newtons (N) using a force meter or spring balance
- A forcemeter is made of a spring with a hook attached to it. The large the force applied, the longer the spring stretches and the greater the reading. The scale of a forcemeter is in Newtons.
- A spring balance is a type of forcemeter that measures how much a spring stretches when an object is hung from it. This is the object's mass. The mass reading on a spring balance can be converted to weight by multiplying the mass by the force of gravity(about 10N per kg) weight is measured in Newtons



100g = 1N

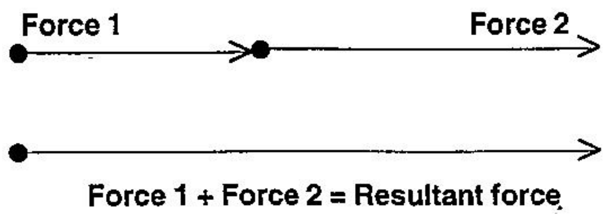
1kg = 10N

BALANCED AND UNBALANCED FORES

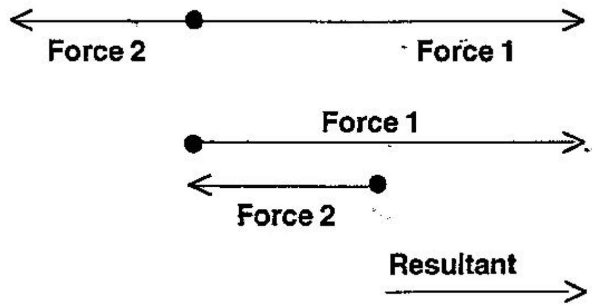
- If two equal sized forces acting in opposite direction are balanced, no movement will occur thus the forces are in equilibrium.
- If two forces are unbalanced then movement will occur in the direction of the larger force.

Resultant force

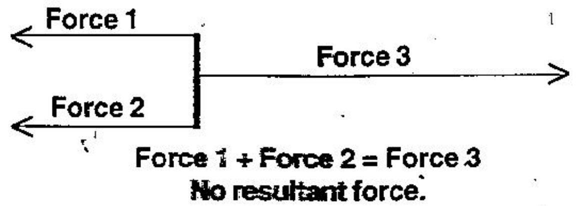
- Is the sum or difference of all forces acting along the same line
    - When forces act along the same line and in the same direction their resultant is the sum of the individual forces



- When forces acting in the same line but in opposite direction their resultant is the difference of the individual forces.



- The resultant force on a body causes movement in its direction. If the resultant is zero, the body is at equilibrium and does not move i.e. equal force in opposite directions.



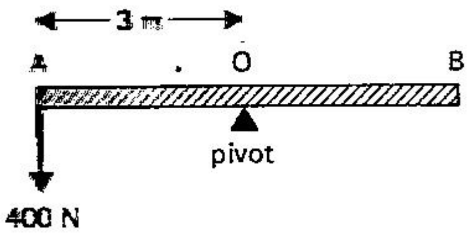
- Direction in which the force is acting is shown by arrows and if drawn to scale they indicate the size of the force.
- The resultant force on a body cause movement in its direction

Example 1 A small cart is pulled by 2 oxen pulling with 200N. The force of friction between the cart and the ground is 50N and between the wheel and axle is 75N. Find the resultant force.

Example 2 A wheelbarrow is pushed along a force of 150N. The force of friction between wheel and the ground is 30N. Draw a diagram to show the forces acting and hence find the resultant force.

MOMENT OF A FORCE

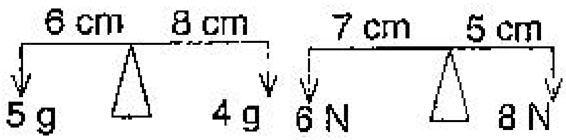
- Moment is the turning effect of a force about a fulcrum (pivot)
- It depends on both the size of the force and how far it is applied from the pivot
- It is measured by multiplying the force by the perpendicular distance of the line of action of the force to the fulcrum.
- In the diagram below, the moment, M, of the force P is given by $\mathbf{M = Fx}$



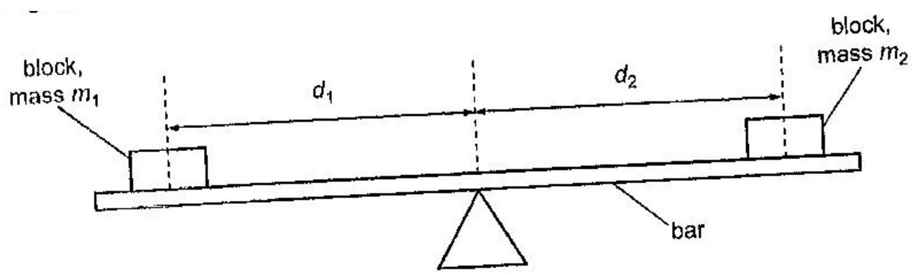
$$Moment\ of\ a\ force = force\ applied\ \times\ distance\ from\ turning\ point$$

- Moment is measured in Newton metres (Nm)
- The greater the force and the distance from the fulcrum, then the greater the turning effect.

Examples Calculate the clockwise and the anticlockwise moments for each of the following examples below.



The principle of Moments

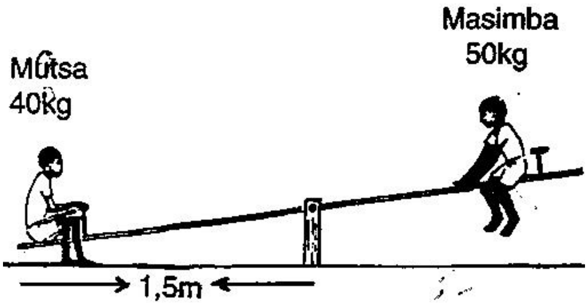


- The principle is about objects balancing when the moments of all the forces acting are balanced.

- The principle states that for a body to be in equilibrium (balanced), the sum of the moments about any point is zero i.e. clockwise moment = the anticlockwise moments

$$F\ x\ d\ (anticlockwise) = F\ x\ d\ (clockwise)$$

Example 1 Mutsa who has a mass of 40kg, sits at one end of a see-saw that is 3m long. The see-saw pivots at its centre point. Where must Masimba, who has a mass of 50kg, sit to balance Mutsa.



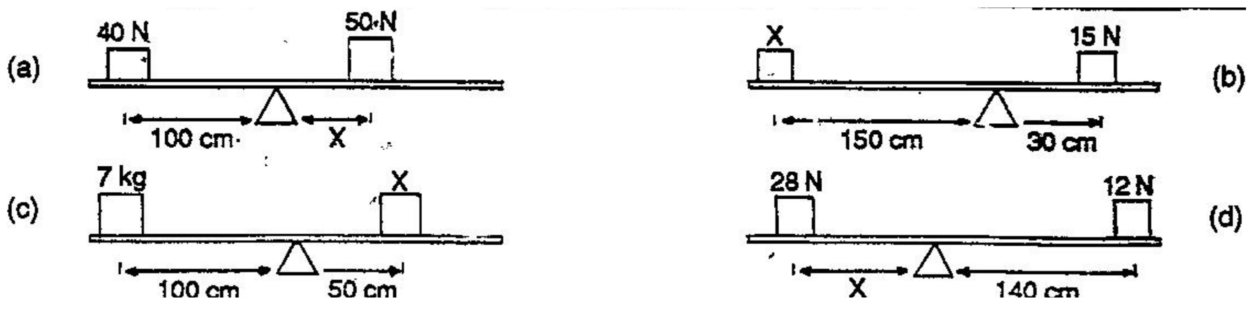
By the law of moments,

Clockwise moments = anticlockwise moments

$\therefore$ 40kg $\times$ 1.5m $=$ 50kg $\times$ Xm

$\therefore$ X $= \frac{40\times1.5}{50} = 1.2$m

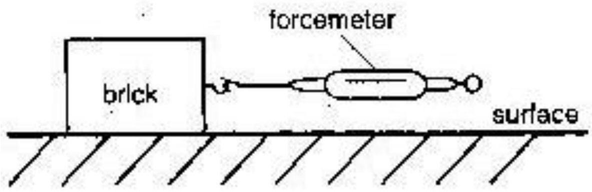
Example 2 Use the principle of moments to calculate the unknown quantities in examples below. In all cases, the beam is balanced.



FRICTION

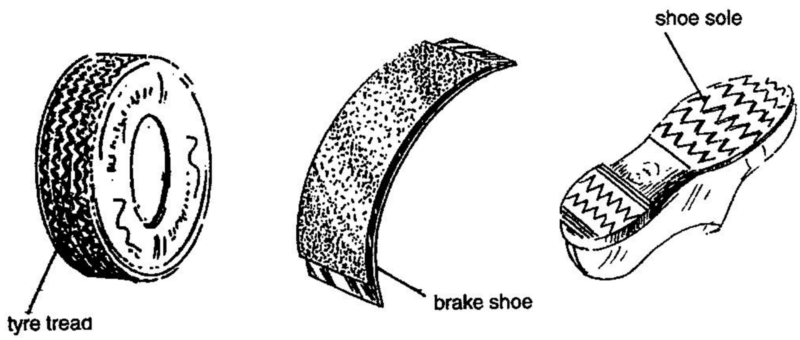
- Friction is a force that opposes motion (movement) when two surfaces are in contact with each other. Friction occurs between two objects that are moving against each other
- An object remains stationary if frictional force is greater than pushing force
- An object only moves if pushing or pulling force is greater than friction.

Measuring friction



- Tie a string around a brick and hook the forcemeter to the string. Use the string to pull the brick.
- Start with a small force and continue to increase the force, noting the reading on the scale of the forcemeter as you increase the pulling force. Record the maximum force reached before the brick starts moving.
- Continue to pull the brick so that it slides with minimum force. Record the reading on the forcemeter while the block is moving.
- As the brick is pulled forward, it didn’t start moving as soon as it is started pulling. Another force acting backwards occurs at the same time i.e. friction which exists between two surfaces that are in contact.
- This opposing force has a maximum size for any two surfaces in contact such that when this size is exceeded, the object starts moving.
- The maximum force recorded by the forcemeter just before the brick starts moving is called static friction.
- As soon as the brick starts moving a smaller force is required to keep brick moving.

Application of frictional forces



Nature of surface – the rougher the surface, the greater the friction it will cause.

Road surfaces – are rough to improve friction

Car breaking systems – the force of the foot on the brake pedal is multiplied by the breaking system. There is a large force at the brake pads against the wheel and the force of friction stops the car.

Tyre threads – the threads on tyres provide a rough surface to improve friction between the wheel and the road surface. This improve grip on the road and prevents skidding.

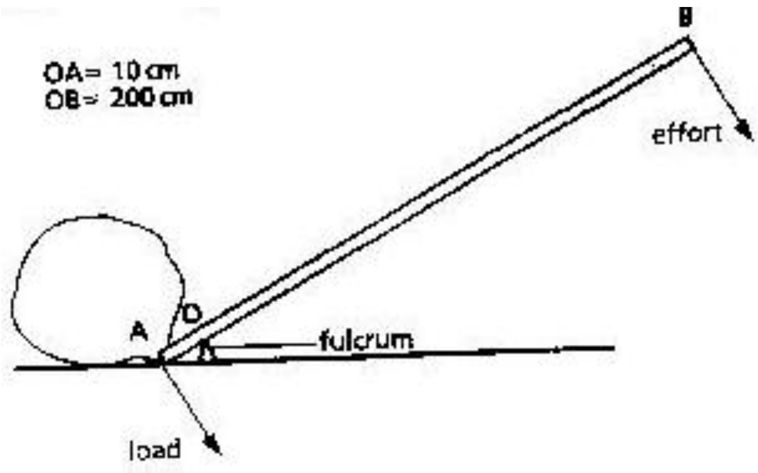
Shoe soles – are rough to increase friction and give a better grip

- All these increase the roughness of surfaces to control motion and remain safe
- Friction can be reduced by
  - Lubricating
  - Use of rollers
  - Use of ball bearings
  - Polishing surfaces
  - Adding wheels

SIMPLE MACHINES

- A machine is a device that helps us to do work easier e.g. levers, pulleys, inclined planes, gears, wheel and axle.
- A machine is anything that magnifies a small force into a large force that is able to do work.
- Machines are energy convertors

Levers

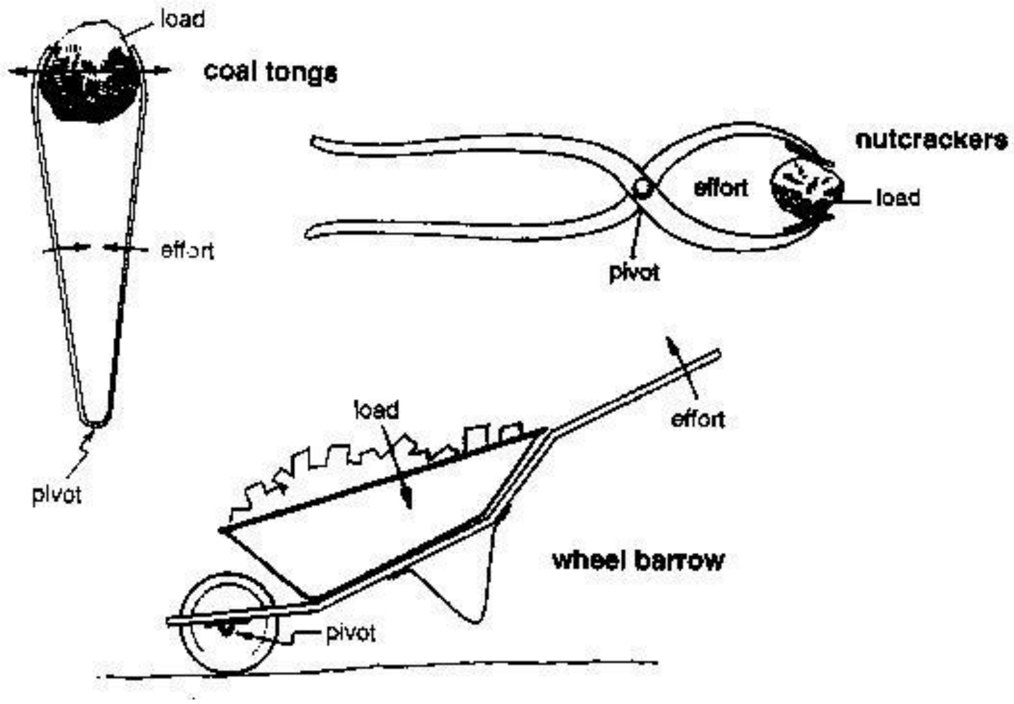


- A lever is a device which can turn about a pivot

- In a working lever a force called the effort is used to overcome a resisting force called the load. The pivotal point is called the fulcrum
- If we use a crow bar to move a heavy boulder, our hands apply the effort at one end of the bar and the load is the force exerted by the boulder on the other end.
- A lever changes the force applied on one end of the lever to an opposite force on the other side

Classes of lever

- First class: fulcrum is between the load and effort e.g. crowbars, scissors, claw hammer, pliers.
- Second class: load is between the fulcrum and effort e.g. wheelbarrow, nutcracker, bottle opener
- Third class levers: effort is between the fulcrum and load e.g. a hoe, fishing rod, tongs, a spade



ENERGY

EFFECTS OF ENERGY

- Energy is the ability to do work
- Effects of energy are things that can be seen as a result of energy e.g.
    - Heating magnesium – burns brightly i.e. gives off much light
    - Squashing a spring – springs changes shape i.e. becomes shorter
    - Pulling elastic band and pluck it – elastic vibrates and make sound
    - Bouncing a ball – ball changes direction i.e. up and down
    - Blowing a whistle – sound comes out
    - Closing switch on an electric motor – motor turns
    - Clapping hands, sound is produced
    - Lighting a torch, chemical energy is converted to light energy
    - Raising a brick, chemical energy is converted to kinetic energy and potential energy
    - Car engines release stored energy and convert it into motion

Sources of Energy

- A source of energy can supply the energy that can be used to do work.
- Source of energy include
    1. Renewable sources i.e. wind, bio-fuels, solar, hydropower
    2. Non-renewable i.e. fossil fuels, nuclear energy

FORMS OF ENERGY

Kinetic energy

- Energy possessed by a body by virtue of its motion e.g. when kicking or bouncing a ball, energy is used to move the ball. The ball has kinetic energy. All moving objects have kinetic energy. The faster it moves the more kinetic energy it has.

Heat energy

- Heat energy is also called thermal energy and is the final fate of other forms of energy. Heat energy causes a change in the internal energy. It is transferred by conduction, convection or radiation. The greater the heat energy of a substance is, the higher its temperature.

Electrical energy

- Energy formed when electrons flow through a conductor. It can be changed into many other forms like heat and light energy. It is produced by energy transfers at power stations and in batteries

Chemical energy

- Chemical energy is energy released if a chemical reaction takes place**.** The energy of food is released by chemical reactions in our bodies during respiration. Fuels cause energy release when they are burnt. Batteries are compact sources of chemical energy, which in use is converted to electrical energy.

Potential energy

- Potential energy is the energy a body has because of its position or condition e.g. water stored in a reservoir has potential energy stored in the form of gravitational potential energy. It is therefore stored up energy that can be released to do work
- There are different types of potential energy i.e.
    - *Gravitational potential energy* – is the energy stored in a body due to its relative position to the earth. For example, water stored behind a dam wall, it has more potential energy than water at lower level. When released, the water loses potential and gains kinetic which can used to drive machinery which generate electricity
    - *Elastic potential energy* – is the energy stored by a stretched or squeezed body. For example, a stretched bow or catapult has elastic potential energy which can be used kinetic energy to a missile. Similarly, a wound up spring can drive a watch. The more an object is stretched or squeezed, the greater its elastic potential energy.
    - *Chemical potential energy* – is the energy within objects that will be converted if a chemical reaction takes place. Food and fuels are stores of chemical

energy. Foods release energy as a result of chemical reactions in the body. When fuels are burnt they release energy mainly as heat. The heat is used to drive an engine.

Light energy

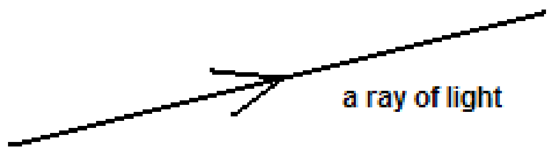
- Light energy is the energy that enables us to see and is important for green plants to make food by photosynthesis. Main source is the sun.

Sources of light energy

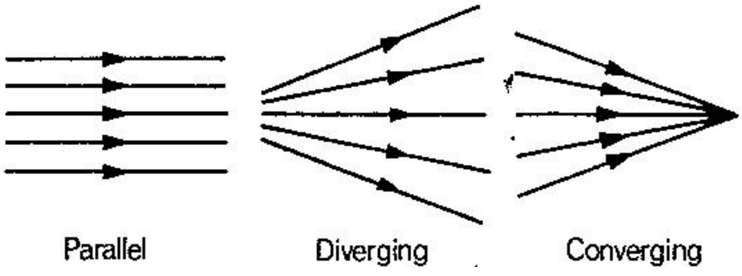
- Sun
- Moon
- Fire
- Electric bulb
- Candle
- LEDs

Transmission of Light

- A ray of light is a narrow stream of light and is shown by straight lines with arrows showing the direction in which the light is travelling.

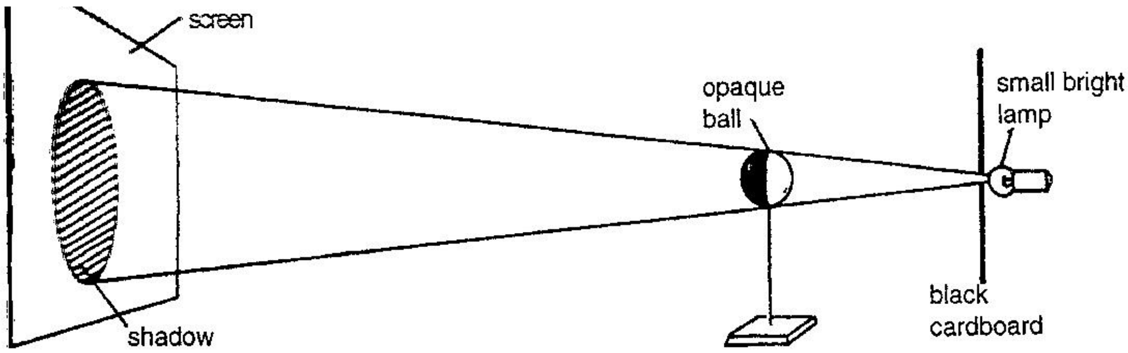


- A beam of light is a stream of light usually represented by a number of rays. A beam can be parallel, divergent or convergent as shown below



Production of shadows

- A shadow is an area in which light does not fall
- Shadows are formed because some (opaque) objects do not allow light to pass through them and that light travel in a straight lines. Sharpness of the shadow depends on the size of the light source.
- If light is made to pass through a narrow hole, a straight ray of light is produced. This ray has defined edges. This is because light travels in a straight.
- If the light is blocked by a figure with sharp edges, a sharp shadow of the object with well defined edges is cast onto a screen.



Sound energy

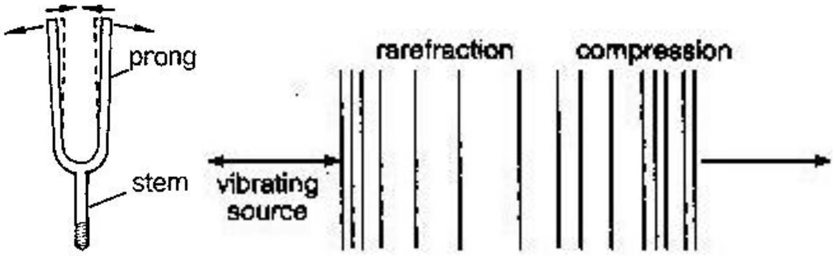
- Sound energy is energy produced when an object or substance vibrates and it moves through materials as sound waves.

Production of sound energy

- Sound is produced by vibrations.
- Vibrations are small but fast backward and forward movements.
- If an elastic band is plucked, it will be seen to vibrate and it will produce a humming sound. The elastic band represents the tissue called the vocal chords found in the throat which vibrate to produce sound.
- Musical instruments make sound waves when part of them or the air inside them vibrates

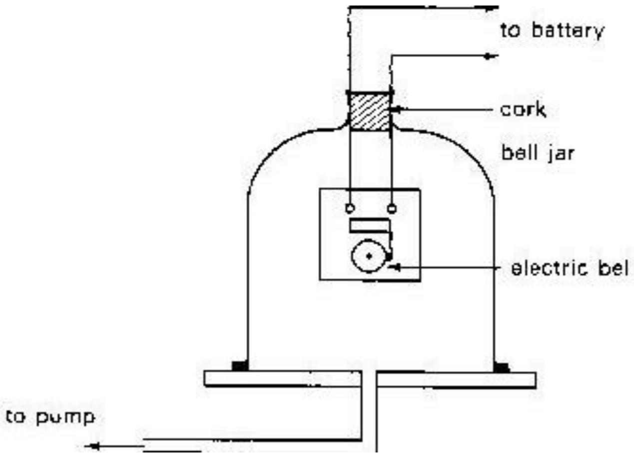
Transmission of sound energy

- Sound waves are mechanical in nature in that they require a medium through which to move which is usually air, liquids or solids.
- Sound are transmitted when particles of a medium vibrate back and forth in the same direction
- When these molecules vibrate, they in turn cause the molecules around them to vibrate causing the sound to be transmitted
- Diagram shows how a vibrating tuning-fork sends out a sound wave. When the tuning fork moves to the right it compresses the air particles together. This disturbance is then transmitted from particle to particle through the air, with the result that a pulse of compression moves outwards.



- A series of compressions and rarefaction moves sound forwards, but the particles of the medium only move backwards and forwards. This is why sound needs a medium in order to be transmitted.

Experiment to show that sound requires a material medium for transmission



- The bell-jar experiment can be used to demonstrate that sound waves cannot pass through a vacuum.
- Before starting the pump, switch on the electric bell. You can see the movement of the striker and hear the sound at the same time.

- When the vacuum pump starts, the sound becomes fainter and fainter with time. Finally it becomes so soft that you cannot hear it any more although the striker can still be seen hitting the gong. If you let air into the bell jar, sound is heard again. This shows that without a medium sound cannot be transmitted.
- Care must be taken that the bell should not touch the glass and the connecting wires used must be thin to prevent any sound waves from being transmitted through them to the outside of the jar as the striker vibrates vigorously

ENERGY CONVERSIONS

Conservation of energy

- The law of conservation of energy states that the total energy of a closed system is constant i.e. energy is neither lost nor gained but simply transformed from one form to another.
- The total energy of a closed system will be same before an interaction as after it
- When energy is transformed from one form to another either work is done or energy is used.

Energy convertors

- Torch
- Dynamo
- Catapult
- Solar panel
- Bulb
- Green plants

Energy chains

- **Energy chain** is a diagram that shows energy conversion.
- Energy can be converted from one form to another

Green plants solar energy from the sun → chemical energy in carbohydrates

Catapult chemical energy → kinetic energy → potential energy → kinetic energy

Dynamo kinetic energy of dynamo → electrical energy → light energy

Bulb electrical energy $\rightarrow$ heat and light energy

Torch chemical energy from cells $\rightarrow$ electrical energy $\rightarrow$ light enrgy

Solar panel solar energy $\rightarrow$ electrical $\rightarrow$ chemical energy in cells

WORK AND ENERGY

Energy

- Is the capacity to do work
- SI units is the Joule (J)
- The amount of work done should be equal to the amount of energy used.

Work

- Work is done when a force moves. It is a measure of the amount of energy transferred
- The amount of work done depends on the size of the force applied distance and distance it moves
- Work is a measure of energy expended in moving an object.

$$\text{Energy used} = \text{Work done}$$

$$work\ done\ = force\ \times\ distance\ moved\ in\ the\ direction\ of\ force$$

- The unit of work is the joule (J) or Newton metre (Nm)
- One joule it is the work done when a force of one Newton (N) moves through one metre
- For example; if you have to pull with a force of 50N to move a crate steadily 3m in the direction of the force, the work done is

$$50N \times 3m = 150N = 150J$$

Example 1 How much work is done when a weight lifter lifts eight 100N boxes through a height of 0.50m?

Example 2 A girl lifts a pile of books with a total mass of 5kg through a vertical distance of 0.75m. How much work does she do?

MAGNETISM

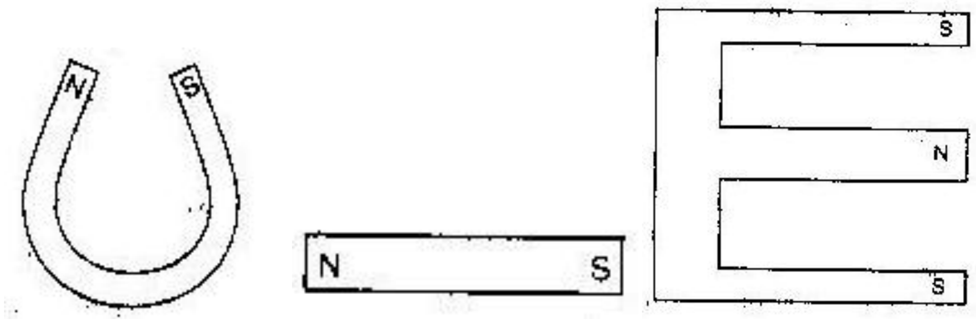
MAGNETS

- Magnetism refers to magnetic force caused by the unique properties of certain materials.
- Are usually made of metal iron or steel

Properties of magnets

1. Can pick magnetic materials
2. Has two poles i.e. north and south pole
3. It has a magnetic field
4. Exerts repulsive and attractive force

Types of magnets



Bar magnet – consist of a straight bar of magnetic material with one end being north and the other south pole. This shape may result in a weak magnetic force as the magnetic force is weak on the sides and is concentrated at the ends of the bar magnet. Can be used in refrigerator doors

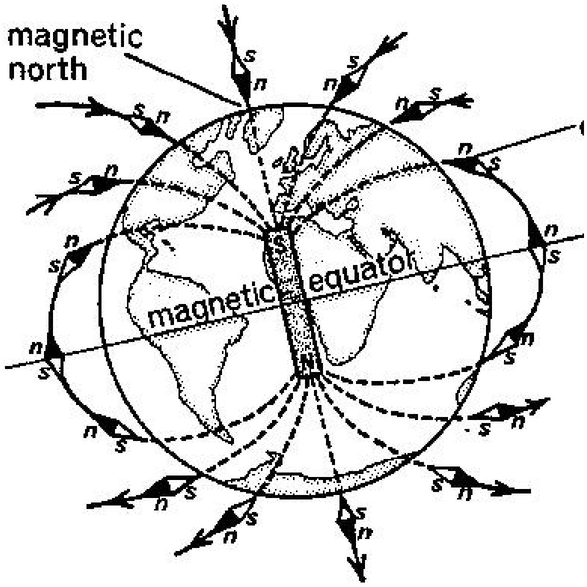
Horse shoe magnet – are bar magnets that bent in a U-shape. The N and S poles point in the same direction resulting in a strong magnetic force that is stronger around both poles. Magnetic force of both poles is felt. It can be used for lifting any metal objects depending in size and strength.

C magnets – are shaped like C, in an arc. They are used to create a magnetic force in a motor. Used in washing machines, fridges, speakers, air conditioners, generators etc

E (Electromagnets) – magnetic field is caused by an electric current. It consists of a coil wire wound around a soft iron core. The wire is connected to a power source. When current flows through, it causes a magnetic field that magnetises the iron core

The earth as a magnet

- If lines of force are plotted on a sheet of paper with no magnets near, a set of parallel straight lines are obtained. They run roughly from S to N geographically and represent a small part of the earth's magnetic field in a horizontal plane.



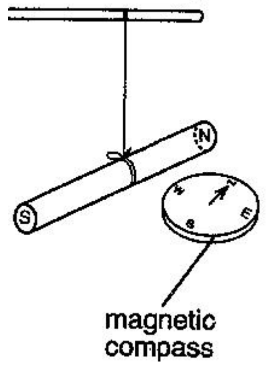
- The earth has opposite poles i.e. the north-pole at the top of the globe and south-pole at the bottom of the globe. It is surrounded by a magnetic field.
- The direction of the earth's magnetic field lines is from the magnetic north pole to the magnetic South Pole.
- The earth's magnetic pole and geographical poles are opposite. The earth's magnetic south pole is where the geographical north pole is and the earth's magnetic south pole is where geographical north pole is

Magnetic from non magnetic materials

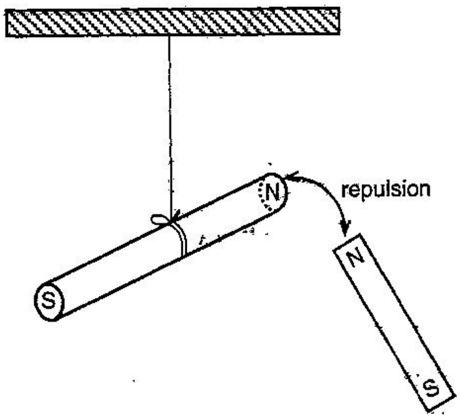
- Magnets can attract some objects. All materials attracted by magnets are called magnetic materials e.g. iron, nickel, cobalt
- All materials that are not attracted by a magnet are called non magnetic materials e.g. wood, rubber, plastic, glass, copper, aluminium.

Magnetic poles

- Magnets have two poles i.e. north and south poles. Magnetism is concentrated around the poles (ends) of a magnet.
- Poles of a freely suspended magnet that always rest pointing towards the north is the north pole(N) while the one that points to the south is the south pole (S)



- Magnetic force acts mainly at the poles of a magnet
- To determine the polarity of bar magnets;
    - Take one bar magnet and hang it freely on a retort stand
    - Bring one pole of the second magnet close to the hanging magnet. Observe what happens.
    - Turn the second magnet so that the poles face the opposite and hold it close to the hanging magnet. Observe what happens.

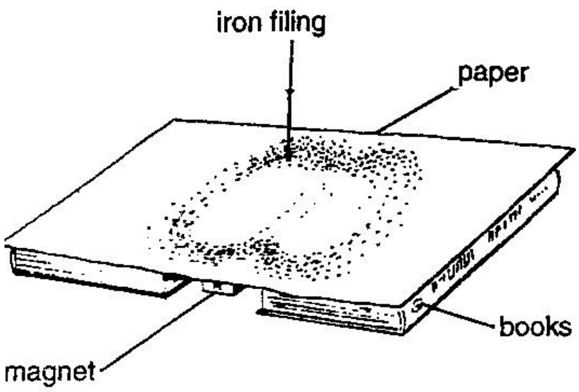


- If the N pole of a magnet is brought near the N pole of another magnet, repulsion occurs. Two S (south seeking) poles also repel. By contrast, N and S poles always attract.
- The law of magnetic poles summarises these facts and states:

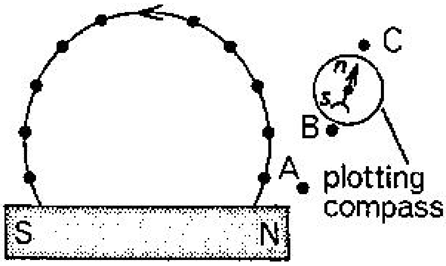
  Like poles repel, unlike poles attract.
- The force between magnetic poles decreases as their separation increases.
- To test whether an object is a magnet, bring one end of this object towards one end of a suspended bar magnet. If repulsion occurs, the object is a magnet since repulsion occurs between like poles
- A given sample is magnetic only if one of its ends repels a magnet. Attraction is not used to verify a magnet because any magnetic material is attracted even if it is not a magnet.

Magnetic fields

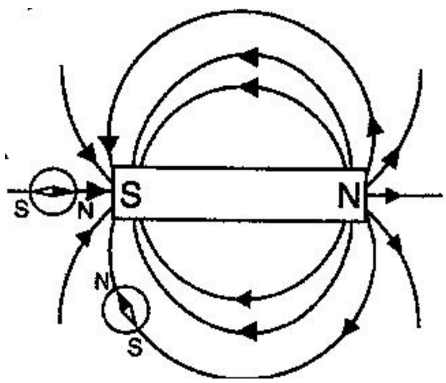
- A magnetic field is the region where magnetic force is exerted on any magnetic object placed within the influence of the field.
- Magnetic field lines show the direction and strength of the magnetic field around a magnet. The direction of magnetic field is always from North Pole to South Pole.
- Magnetic force is strongest at the poles of a bar magnet where field lines are most concentrated that is at the poles.
- To show the pattern of a magnetic field around a bar magnet, place a sheet of paper on top of a bar magnet and sprinkle iron filings lightly and evenly on to the paper. Tap the paper gently with a pencil and the filings should form patterns showing the lines of force. The iron filings will line up along the magnetic field lines.



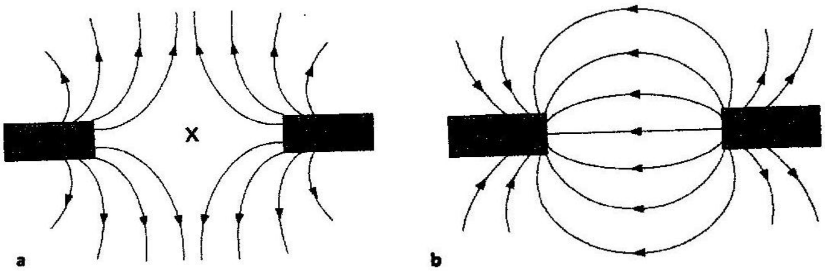
- The plotting compass can also be used to plot the combined magnetic field due to neighbouring magnets or around a bar magnet.
- Lay a bar magnet on a sheet of paper. Place the plotting compass at appoint such as A, near one pole of the magnet (north pole)
- Mark the position of the poles (n; s) of the compass by pencil dots. B,A
- Move the compass so that the pole s is exactly over B, mark the new position of n by dot C
- Continue this process until the other pole of the bar magnet is reached. Join the dots to give one line of force and show its direction by putting an arrow on it. By convention, the field direction draws north to south.



- Plot other lines by starting at different points around a bar magnet as shown below



- Magnetic field gets weaker as the distance from the magnet increases – the lines are further apart
- The field lines also help explain what happens as two magnets are brought together – the field lines will interact with each other giving either repulsion or attraction, depending on the direction of the interacting field lines

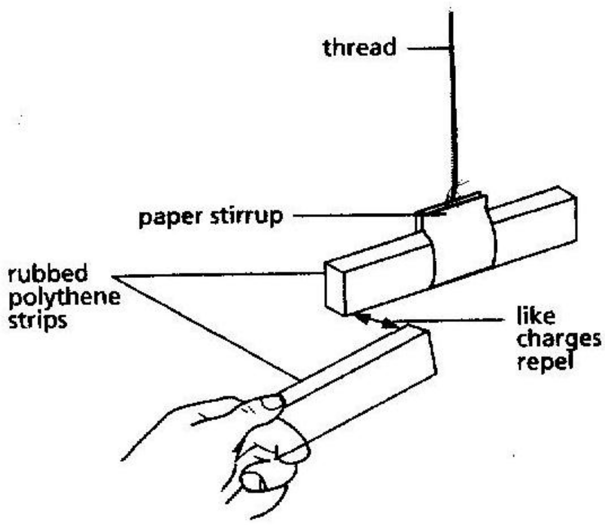


ELECTRICITY

STATIC ELECTRICITY

Charges

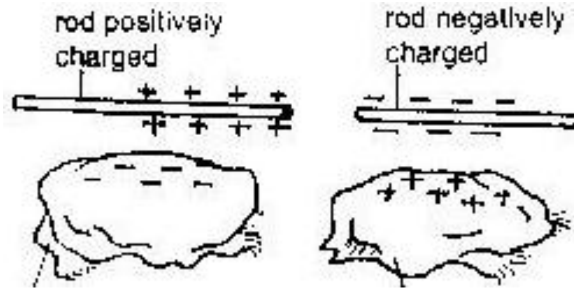
- When a strip of polythene is rubbed with a cloth it becomes charged. If it is hung up and another rubbed polythene strip is brought near, repulsion occurs. Attraction occurs when a rubbed strip of Perspex is brought near.



- This shows that there are two kinds of electric charges i.e.
  - **positive** (+) (on the perspex rod)
  - **negative** (–) (on the polythene rod)
- It shows that like charges repel, while unlike charges attract.

The production of charges

- Atoms are made up of positively charged protons, negatively charged electrons and neutrons that have no charge.
- If an object gains electrons, it will become more negatively charged and if it loses electrons, it will become more positively charged.
- Rubbing a rod using a woollen cloth produces an electrical charge. The production of charges by rubbing can be explained by supposing that electrons are transferred from one material to the other. Either the electrons are removed from the rod or electrons are added onto the rod.



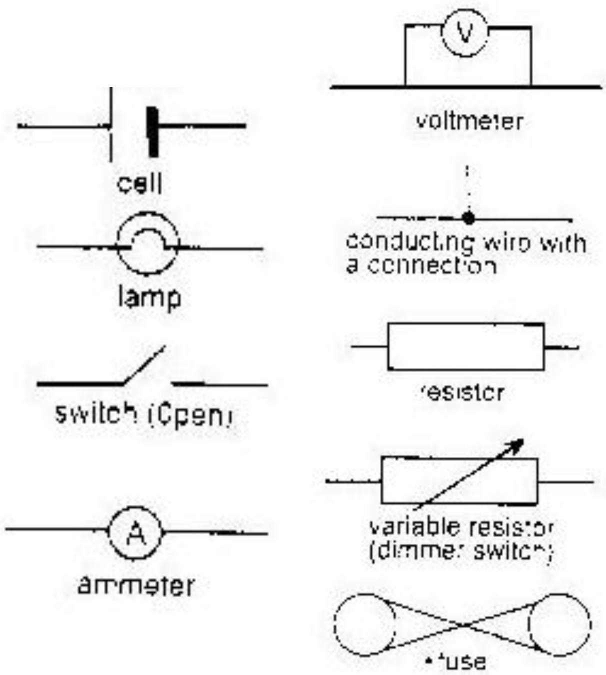
- Rubbing polythene with a cloth makes it become negatively charged i.e. electrons are added onto the rod
- Rubbing perspex with a cloth makes it become positively charged i.e. electrons are removed from the rod.
    - *Note that it is only electrons which move; the protons remain fixed in the nucleus.*

CURRENT ELECTRICITY

Simple d.c. circuits

- It is important to draw simple and clear diagrams. The following diagrams gives a list of special symbols that are used to represent common devices that are usually employed in electric circuits

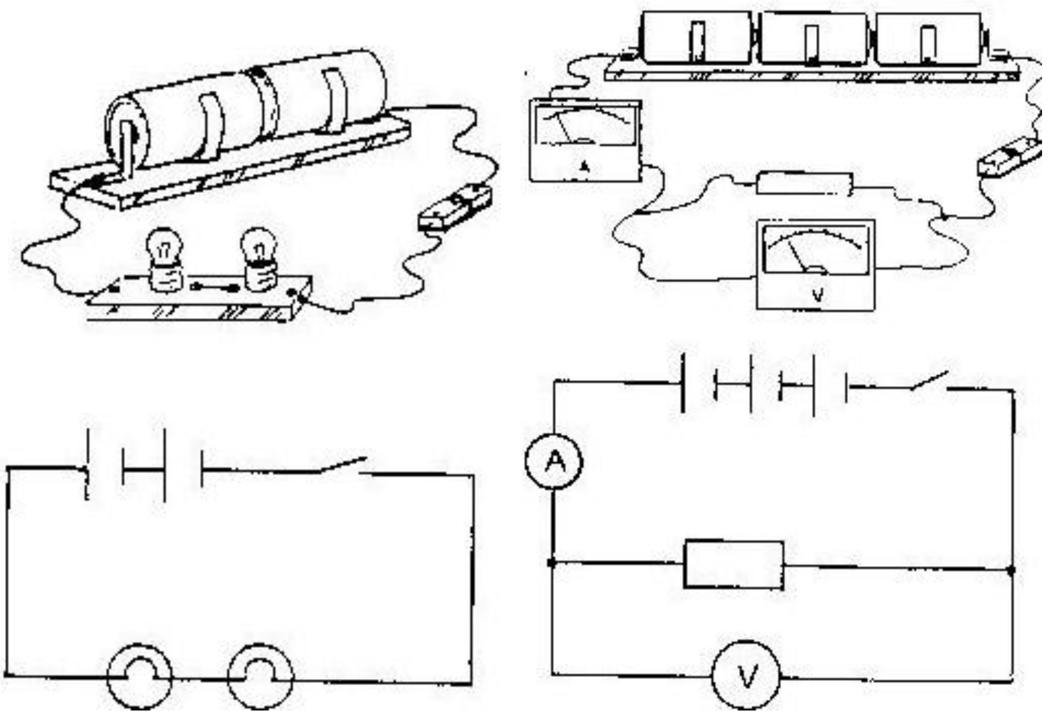
Circuit symbols



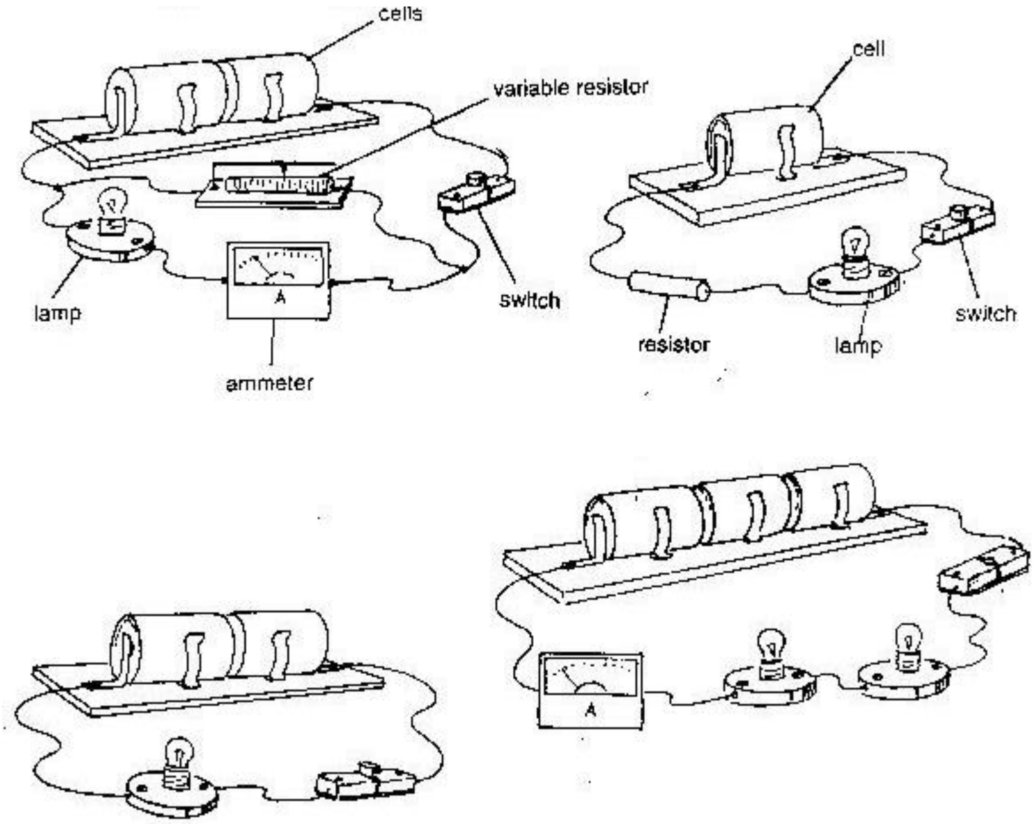
Components of direct current circuit

- The cell is the source of electrical energy. The longer line is the positive terminal and shorter and thicker line is the negative terminal of the cell.
- The switch breaks or completes an electric circuit, stopping or starting the current flow. When switch is open, circuit is incomplete and current does not flow. When the switch is closed the current flows since the circuit is complete.
- A resistor is a device that offers opposition to the flow of current. It restricts the flow of current in circuits, slowing the current down.
- A light bulb is a resistor that changes electrical energy into light and heat energy.
- A fuse is a device that protects appliances and users from electrical shocks
- An ammeter is used to measure the amount of current
- A voltmeter is used to measure voltage of the circuit
- A variable resistor regulates or controls the current in a circuit i.e. it can increase or decrease the amount of current.
- Connecting wires provide path for electric current
- *Every circuit has a power source, switch and a load (all components that use the electrical energy and convert it to other forms)*

Representing some circuit diagrams using electric symbols



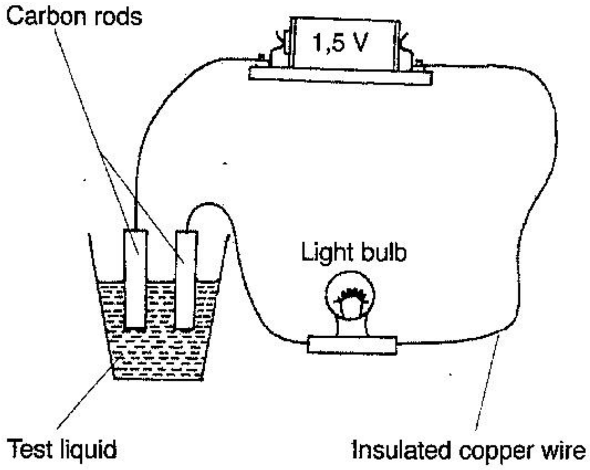
Draw a standard circuit diagram using symbols for each of the circuits shown below

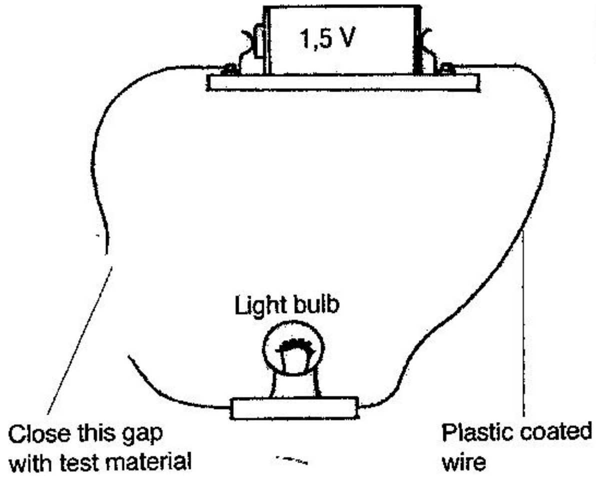


CONDUCTORS AND INSULATORS

- Conductors are materials that allow electricity to flow through it e.g. copper, carbon, and salt water.
- Insulators are materials that does not conduct an electric current e.g. rubber, plastic, wood, glass, pure water
- A solution that conducts electricity is called an electrolyte. The solution is chemically changed by the current
- Conductors are generally used for transmission of electricity.
- Insulators generally protects people from electrocution
- Most metals are conductors while non metals are insulators except for graphite (carbon).
- Gases are also very poor conductors of electricity

Experiment on conductivity of different materials





- Make a simple circuit using a battery connected to two wires and a light bulb as shown in the diagram.
- Connect different materials using crocodile clips to find out which ones conduct or do not conduct electricity. If they do conduct electricity the light bulb will light up.
- Record results in the table below

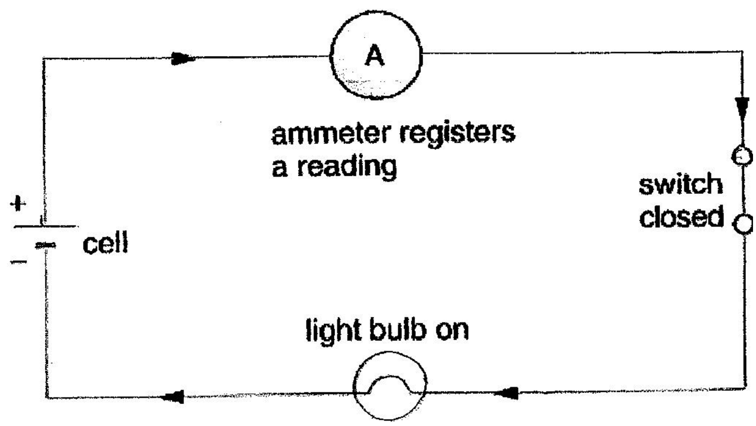
Material	Observation	Conclusion

CURRENT, VOLTAGE AND POWER

Current

- Is the flow of charge in an electric circuit
- An ammeter is used to measure current
- The unit of current is the ampere (A) or amps

Measuring current



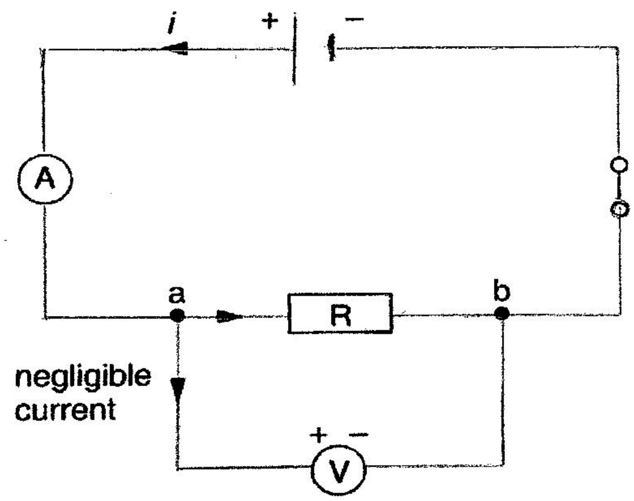
- An ammeter is always placed in series with the resistance or other circuit components through which the current is to be measured.
- Ammeters should therefore have low resistance compared with that of the rest of the circuit, so that they do not introduce unwanted resistance

Voltage

- Is the measure of the ability of electrical energy to do work.
- It is measured in volts (v) using a voltmeter
- It is supplied by the cell or battery

Measuring voltage

- A voltmeter is always placed in parallel with the resistance or apparatus across which the potential difference has to be measured.
- Voltmeter ought therefore to have a high resistance compared with the resistance across which the voltage is to be measured, so that they take a comparatively negligible current, and so disturb the circuit as little as possible.



Power

- Electrical power ***is*** the rate at which electrical energy is changed to other forms of energy in an electric circuit.
- Power is measured in watts (W).

$$power = voltage\ x\ current$$

$$P = VI$$

Example 1 What is the power of an electric light bulb if it is drawing a current of 0.25A from the mains electricity supply of 240V?

Example 2 What current is drawn by a 1.5kW heater which operates on a 240V mains supply?

Example 3 What voltage is needed for a 0.5A current to pass through a 100W light bulb